

**P.P. NAPROXEN NA 275MG
P.P. NAPROXEN NA 550MG
TABLET**

Composition:

P.P. NAPROXEN NA 275MG: Each tablet contains naproxen sodium 275mg.
Light blue coloured oblong shaped film-coated tablet with score line one side and plain on the other side.
P.P. NAPROXEN NA 550MG: Each tablet contains naproxen sodium 550mg.
Blue- coloured oblong shaped film-coated tablet with score line on one side.

Mechanism of Action:

Naproxen is a non-steroidal, anti-inflammatory drug (NSAID) of the acryacetic acid group. The sodium salt of naproxen is a more rapidly absorbed form of naproxen. The naproxen anion inhibits cyclooxygenases which are involved in the biosynthesis of prostaglandins.

Pharmacokinetics:

Naproxen sodium is rapidly and completely absorbed from the gastrointestinal tract with an in vivo bioavailability of 95%. Peak plasma concentrations are achieved in 1 to 2 hours after ingestion of a dose of naproxen sodium, with steady state levels reached in 4 to 5 days. At therapeutic levels, naproxen is more than 99% bound to albumin. Plasma levels of naproxen increase proportionally with dose up to about 500mg per day; at higher doses, there is an increase in clearance caused by saturation of plasma protein binding.

Naproxen has a plasma half-life of about 13 hours. Approximately 95% of a given dose is excreted via the kidneys as naproxen and its metabolite, 6-O-desmethylnaproxen and their conjugates. Less than 3% of a dose is found in the faeces. Naproxen crosses the placenta and is excreted in breast milk.

Indications:

1. In the treatment of musculoskeletal and joint disorders such as ankylosing spondylitis, osteoarthritis, rheumatoid arthritis and acute gout.
2. In the treatment of mild to moderate pain such as migraine, primary dysmenorrhoea, acute tendonitis and bursitis.

Contraindications:

- Known hypersensitivity to naproxen or any excipient of the drug. It should not be given to patients in whom aspirin or other nonsteroidal anti-inflammatory drugs (NSAIDs) have induced asthma, nasal polyps or other hypersensitivity reactions
- Active peptic ulceration
- Products containing naproxen or naproxen sodium are contraindicated in children under 2 years of age since safety in this age group has not been established.

Dosage and Administration:

Therapy should be started at the lowest recommended dose, especially in the elderly. After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

Adults

Analgesia/dysmenorrhoea/acute musculoskeletal conditions/acute pain states in which there is an inflammatory component:

Because the sodium salt of naproxen is more rapidly absorbed, naproxen is recommended for the management of acute painful conditions when prompt onset of pain relief is desired. The recommended starting dose is 550mg followed by 275mg every 6 - 8 hours as required. A total dose of 1375mg per day should not be exceeded.

Acute gout: The recommended starting dose is 825mg followed by 275mg every 8 hours as needed.

Menorrhagia: The recommended dosage is 825 – 1375mg per day taken in 2 doses on the first day of menstrual bleeding. Thereafter, the total daily dose should not exceed 1100mg.

Migraine: For treatment of acute migraine headache, the dose is 825mg at the first symptom of an impending attack. An additional dose of 275mg to 550mg can be taken throughout the day, if necessary, but not before half an hour after the initial dose.

For prophylaxis of migraine headache, the recommended dosage is 550mg twice daily. If no improvement is seen within 4 – 6 weeks, the drug should be discontinued. A total daily dose of 1375mg per day should not be exceeded.

Elderly

Studies indicate that although the total plasma concentration of naproxen is unchanged, the unbound plasma fraction of naproxen is increased in the elderly. The implication of this finding for naproxen dosing is unknown but it is prudent to use the lowest effective dose. For the effect of reduced elimination in the elderly, refer to the section *Use in patients with impaired renal function*.

Children

Naproxen is not recommended for use in children under sixteen years of age.

Route of Administration:

Oral administration

Warning and precautions:

Episodes of gastrointestinal bleeding have been reported in patients with naproxen or naproxen sodium therapy. Naproxen should therefore be given under close supervision to patients with a history of gastrointestinal disease.

Serious gastrointestinal adverse reactions can occur, at any time in patients on therapy with NSAIDs. The risk of their occurrence does not seem to change with duration of therapy. Studies to date have not identified any subset of patients not at risk of developing peptic ulcer and bleeding, however, elderly and debilitated patients tolerate gastrointestinal ulceration or bleeding less well than others. Most of the serious gastrointestinal events associated with NSAIDs occurred in this patient population.

The antipyretic and anti-inflammatory activities of naproxen may reduce fever and inflammation, thus diminishing their usefulness as diagnostic signs.

Bronchospasm may be precipitated in patients suffering from, or with a history of, bronchial asthma or allergic disease.

Sporadic abnormalities in laboratory tests (e.g. liver function tests) have occurred in patients on naproxen or naproxen sodium therapy, but no definite trend was seen in any test indicating toxicity.

Naproxen decreases platelet aggregation and prolongs bleeding time. This effect should be kept in mind when bleeding times are determined.

Naproxen may cause mild peripheral oedema. Although sodium retention has not been reported in metabolic studies, it is possible that patients with questionable or compromised cardiac function may be at a greater risk when taking naproxen.

Each P.P. NAPROXEN NA 275MG or P.P. NAPROXEN NA 550MG tablet contains 25mg or 50mg of sodium (about 1 mEq per each 250mg of naproxen). This should be considered in patients whose overall intake of sodium must be severely restricted.

Use in patients with impaired renal function: As naproxen is eliminated to a large extent (95%) by urinary excretion via glomerular filtration, it should be used with great caution in patients with impaired renal function and the monitoring of serum creatinine and/or creatinine clearance is advised in these patients. Caution should be used if the naproxen is given to patients with creatinine clearance of less than 20ml/minute.

Certain patients, specifically those whose renal blood flow is compromised, such as in extracellular volume depletion, cirrhosis of the liver, sodium restriction, congestive heart failure, and pre-existing renal disease, should have renal function assessed before and during naproxen therapy. Some elderly patients in whom impaired renal function may be expected, as well as patients using diuretics, may also fall within this category. A reduction in daily dosage should be considered to avoid the possibility of excessive accumulation of naproxen metabolites in these patients.

Use in patients with impaired liver function: Chronic alcoholic liver disease and probably also other forms of cirrhosis reduce the total plasma concentration of naproxen, but the plasma concentration of unbound naproxen is increased. The implication of this finding for naproxen dosing is unknown but it is prudent to use the lowest effective dose.

Haematological

Patients who have coagulation disorders or are receiving drug therapy that interferes with haemostasis should be carefully observed if naproxen-containing products are administered.

Patients at high risk of bleeding or those on full anti-coagulation therapy (e.g. dicoumarol derivatives) may be at increased risk of bleeding if given naproxen-containing products concurrently.

Anaphylactic (anaphylactoid) reactions

Hypersensitivity reactions may occur in susceptible individuals. Anaphylactic (anaphylactoid) reactions may occur both in patients with and without a history of hypersensitivity or exposure to aspirin, other NSAIDs or naproxen-containing products. They may also occur in individuals with a history of angioedema, bronchospastic reactivity (e.g. asthma), rhinitis and nasal polyps.

Anaphylactoid reactions, like anaphylaxis, may have a fatal outcome.

Steroids

If steroid dosage is reduced or eliminated during therapy, the steroid dosage should be reduced slowly and the patients must be observed closely for any evidence of adverse effects, including adrenal insufficiency and exacerbation of symptoms of arthritis.

Ocular effects

Studies have not shown changes in the eye attributable to naproxen administration. In rare cases, adverse ocular disorders including papillitis, retrobulbar optic neuritis and papilledema, have been reported in users of NSAIDs including naproxen, although a cause-and-effect relationship cannot be established; accordingly, patients who develop visual disturbances during treatment with naproxen-containing products should have an ophthalmological examination.

Combination with other NSAIDs

The combination of naproxen-containing products and other NSAIDs is not recommended, because of the cumulative risks of inducing serious NSAID-related adverse events.

Cardiovascular Thrombotic Events

Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.

There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.

Hypertension

NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.

Heart Failure

Fluid retention and oedema have been observed in some patients taking NSAIDs, there caution is advised in patients with fluid retention or heart failure.

Gastrointestinal Events

All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patients about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Severe Skin Reactions

NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

Warnings

Risk of GI ulceration, bleeding and perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (eg dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (eg alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Pregnancy and lactation:

Naproxen should not be used during pregnancy, unless the potential benefits justify the potential risk to the fetus. Regular use of NSAIDs during the 3rd trimester of pregnancy may cause closure of foetal ductus arteriosus in utero and possibly in persistent pulmonary hypertension of the newborn. The onset and duration of labour may be prolonged. Naproxen is excreted in breast milk at a concentration of about 1% of that found in plasma. Because of the potential for adverse effects of prostaglandin-inhibiting drugs on neonates, use in nursing mothers should be avoided.

Effect on ability to drive or operate machinery:

Patients who experience visual disturbances, drowsiness, dizziness or vertigo should refrain from driving or operate machinery.

Drug Interactions

1. Angiotensin-converting enzyme (ACE) inhibitors, cyclosporin: Concurrent use with naproxen or other NSAIDs increases the risk of nephrotoxicity.
2. Diuretics: The natriuretic effects of frusemide may be inhibited.
3. Antihypertensives : NSAIDs may reduce the antihypertensive efficacy of these drugs due to inhibition of the synthesis of vasodilating prostaglandins.
4. Phenytoin, sulphonylurea drugs: Concurrent use may enhance the effects of these drugs.
5. Probenecid: Probenecid given concurrently increases plasma levels of naproxen and prolongs its half-life.
6. Aspirin: Concomitant administration of naproxen with aspirin is not recommended because naproxen is displaced from its binding sites by aspirin, resulting in lower peak and plasma concentrations.
7. Lithium, methotrexate: Plasma levels and toxicities of these drugs may be enhanced by naproxen

Side-Effects

The most commonly reported adverse effects of naproxen include:

Gastrointestinal : constipation, heartburn, abdominal pain, nausea, dyspepsia, diarrhoea, stomatitis.

Central nervous system : headache, vertigo, lightheadedness, drowsiness, dizziness, insomnia, rarely, aseptic meningitis

Dermatological: pruritus, rashes, urticaria, skin eruptions

Cardiovascular : oedema, dyspnea, palpitations

Haematological : anaemia, thrombocytopenia, neutropenia, eosinophilia, granulocytosis.

Others: thirst, abnormal liver function tests, glomerular nephritis, hematuria, interstitial nephritis, nephrotic syndrome, tinnitus, visual disturbances

Overdosage:

Symptoms of overdose may include drowsiness, heartburn, indigestion, nausea or vomiting. The stomach should be emptied and supportive treatment should be instituted.

Haemodialysis does not decrease the plasma concentration of naproxen.

Storage conditions:

Protect from light.

Store in a dry place below 30°C.

Keep out of reach of children.

Jauhkan dari kanak-kanak

Presentation

Blisters of 10 tablets. 3, 6, 9, 10, 25, 50 or 100 blisters in a box.

Manufactured by:

Noripharma Sdn. Bhd. 200701034604 (792633-A)

Lot 5030, Jalan Teratai,

5 ½ Mile off Jalan Meru,

41050 Klang, Selangor.

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